



Sir Syed University
of Engineering & Technology

Artificial Intelligence Lab

Project Proposal

Supervised by
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Artificial Intelligence Lab Project

BS Computer Science, Batch 2022F

Medical Image Analysis

Group Members:

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Supervisor Approval

Project Proposal

Title: Medical Image Analysis Using Deep Learning for Disease Diagnosis and Treatment Planning

Introduction

Medical imaging is a cornerstone of modern healthcare, enabling early diagnosis, treatment planning, and monitoring of various diseases. However, interpreting large volumes of complex medical images (e.g., X-rays, MRI scans, histopathology slides) is time-consuming and prone to human error. Our AI project aims to develop a deep learning-based solution that automates the analysis of different types of medical images for tasks such as disease diagnosis (e.g., pneumonia from X-rays), tumor detection (e.g., brain/lung tumors from MRI/CT), and treatment assistance (e.g., identifying cancerous tissues from histopathology slides).

This project is important because it can improve diagnostic accuracy, reduce healthcare costs, accelerate clinical workflows, and provide decision support to radiologists and pathologists—ultimately saving lives and enhancing patient care.

Objectives

- Develop a multi-modal deep learning system to analyze different types of medical images (X-ray, MRI, histopathology).
- Detect and classify diseases such as pneumonia, tumors, and cancerous cells across multiple datasets.
- Provide interpretable results through heatmaps and feature maps to assist healthcare professionals.
- Evaluate model performance using standard metrics like accuracy, sensitivity, and AUC.
- Create a prototype diagnostic dashboard (optional) to demonstrate usability in a clinical context.

Background and Literature Review

Recent studies have demonstrated the power of deep learning, especially Convolutional Neural Networks (CNNs), in medical image analysis. For example, **Rajpurkar et al. (2017)** introduced *CheXNet*, a 121-layer CNN that outperformed radiologists in pneumonia detection using chest X-rays. Another study by **Campanella et al. (2019)** applied deep learning to whole-slide histopathology images and achieved diagnostic performance on par with pathologists for prostate and breast cancer.

These studies confirm the viability and effectiveness of AI in improving diagnostic accuracy and efficiency across various types of medical imaging. However, many models are narrowly focused on a single imaging type or disease. Our project aims to address this gap by developing a multi-disease, multi-modality diagnostic system.

Expected Methodology

- **Dataset Collection:** Use publicly available datasets:
 - Chest X-ray (e.g., NIH ChestX-ray14, RSNA Pneumonia)
 - MRI scans (e.g., BraTS for tumors)
 - Histopathology slides (e.g., BreakHis or PCam datasets)
- **Preprocessing:** Resize images, normalize pixel values, data augmentation, and noise removal.
- **Model Architecture:** Build and compare different CNN architectures (e.g., ResNet, VGG, DenseNet).
- **Training & Testing:** Use 80-20 train-test split with cross-validation.
- **Evaluation Metrics:** Accuracy, precision, recall, F1-score, confusion matrix, ROC-AUC.
- **Explainability:** Use Grad-CAM or LIME to generate visual explanations for predictions.
- **Tools:** Python, TensorFlow/Keras, Jupyter Notebooks, OpenCV, Streamlit (optional dashboard)

Tools

- **Languages & Frameworks:** Python, TensorFlow, Keras, PyTorch
- **Libraries:** OpenCV, NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn
- **Platforms:** Jupyter Notebook, Google Colab, Kaggle
- **Optional UI:** Streamlit or Flask for dashboard prototype
- **Hardware:** Google Colab GPU

Timeline

Phase	Task Description	Start Date	End Date	Assigned Member
Phase 1: Research	Literature review, dataset selection	May 25	May 30	Member 1
Phase 2: Preprocessing	Image enhancement, normalization, augmentation	May 31	June 5	Member 2
Phase 3: Model Development	Train CNN models on X-ray, MRI, histopathology datasets	June 6	June 15	Member 3
Phase 4: Evaluation	Test performance, generate explainability maps	June 16	June 20	Member 1
Phase 5: Deployment (opt)	Build dashboard or demo interface for predictions	June 21	June 25	Member 2
Phase 6: Final Report	Write and compile project documentation	June 26	June 29	Member 1 & 3

Conclusion

This project combines AI and healthcare to address the limitations of manual medical image analysis. By applying deep learning techniques to multiple image modalities, our system can assist in diagnosing diseases more accurately and efficiently. The interpretability of predictions also helps gain the trust of healthcare professionals. We urge academic reviewers and stakeholders to support this project for its strong potential to improve diagnostic workflows, patient care, and medical efficiency.

References

1. Rajpurkar, P., Irvin, J., Zhu, K., Yang, B., Mehta, H., Duan, T., ... & Ng, A. Y. (2017). **CheXNet: Radiologist-level pneumonia detection on chest X-rays with deep learning**. arXiv preprint arXiv:1711.05225.
2. Campanella, G., Hanna, M. G., Geneslaw, L., Miraflor, A., Werneck Krauss Silva, V., Busam, K. J., ... & Fuchs, T. J. (2019). **Clinical-grade computational pathology using weakly supervised deep learning on whole slide images**. *Nature Medicine*, 25(8), 1301–1309.